

Computer vision: Computer vision algorithms interpret images and video data from cameras mounted on the vehicle. They identify objects in the car's environment, such as other vehicles, pedestrians, traffic signs, and traffic lights. Table 1 gives a description of these algorithms.

Lidar processing: Lidar (light detection and ranging) sensors are used to create a 3D map of the car's environment. Algorithms are used to process the data collected by these sensors to create a high-resolution map that the car can use for navigation.

Some of the key algorithms used in lidar processing for self-driving cars are listed in Table 2.

By processing the lidar data with these algorithms, self-driving cars can identify and track objects in real-time, create accurate maps of the environment, and precisely locate themselves within that environment.

Path planning: Path planning algorithms are used to determine the optimal route for the car to reach its destination. These algorithms consider factors such as traffic conditions, road conditions, and the location of other

vehicles on the road (Table 3).

Each of these algorithms has its strengths and weaknesses, and the choice of which algorithm to use depends on the specific needs of the self-driving car system and the characteristics of the environment it operates in.

Control: Control algorithms are used to control the car's speed, steering, and braking. These algorithms consider the car's current speed and location, as well as the route planned by the path planning algorithms to make decisions about how to safely navigate the vehicle (Table 4).

Algorithm	Description
Convolutional neural networks (CNNs)	These are deep learning algorithms that analyse visual data and classify objects in the environment. CNNs are commonly used in object detection and recognition tasks. They can learn to recognise complex patterns and shapes, making them effective in identifying objects in real-world environments.
Semantic segmentation	Used to separate the environment into different regions, such as roads, sidewalks, and buildings. This helps the vehicle to understand the layout of the environment and identify objects more accurately. It works by assigning each pixel in an image to a particular class, which can help the vehicle navigate the environment more effectively.
Optical flow	Used to estimate the motion of objects in the environment. This information predicts the trajectory of objects, such as other vehicles or pedestrians. It works by analysing the changes in brightness between video frames to determine the direction and speed of motion. Optical flow is helpful in detecting unexpected changes in movement including sudden stops or changes in direction.
Stereo vision	Uses two cameras to create a 3D representation of the environment. This enables the vehicle to estimate the distance to objects and plan its movements accordingly. It works by analysing the differences between the images captured by two cameras, to calculate the distance to objects in the scene. Stereo vision is useful in detecting obstacles and estimating the size and position of objects in the environment.
Lane detection	Used to identify the lanes on the road and ensures that the vehicle stays within them. This is typically achieved through image processing techniques, such as edge detection and Hough transforms. It works by analysing the edges in an image to identify the boundaries of the lanes. Lane detection is useful in keeping the vehicle in the correct lane and avoiding collisions with other vehicles.
Object detection	Used to identify objects in the environment, such as vehicles, pedestrians, and bicycles. This is typically achieved through machine learning algorithms, such as CNNs. Object detection algorithms can locate objects in an image or video and determine their class, position, and size. They are used in a variety of tasks including collision avoidance and pedestrian detection.
Image recognition	Used to identify objects in the environment based on their appearance. This is typically achieved through machine learning algorithms, such as CNNs. Image recognition algorithms can classify objects in an image or video and determine their class. They are useful in detecting and identifying objects in the environment like road signs and traffic lights.
Object tracking	Used to track the movement of objects in the environment, such as other vehicles and pedestrians. This is typically achieved through a combination of prediction and filtering techniques to track the object's movement. Object tracking algorithms can estimate the object's position and velocity over time and predict its future position. They are useful in a variety of tasks, such as collision avoidance and pedestrian detection.

Table 1: Computer vision algorithms